

CFD 2025-26 – Lab 02

Exercise 1

Let us consider the Stokes problem defined in the rectangular domain $\Omega = (0, 3) \times (0, 1)$

$$\begin{cases} -\Delta \mathbf{u} + \nabla p = \mathbf{0}, & \text{in } \Omega, \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, \\ \mathbf{u} = \mathbf{0}, & \text{on } \Gamma_{D1} = \{0 \leq x \leq 3, y = 0\}, \\ \mathbf{u} = 1\mathbf{i} + 0\mathbf{j}, & \text{on } \Gamma_{D2} = \{0 \leq x \leq 3, y = 1\}, \\ (\nabla \mathbf{u} - pI)\mathbf{n} = \mathbf{0}, & \text{on } \Gamma_N = \{x \in \{0, 3\}, 0 \leq y \leq 1\}. \end{cases}$$

- 1 Verify that $\mathbf{u} = y\mathbf{i}$, $p = 0$ is a solution of the problem.
- 2 Write the weak formulation of the differential problem.
- 3 Using the variational approach of Firedrake, write a solver for the problem, with $h = 0.1$ as mesh discretization parameter and $\mathbb{P}^1/\mathbb{P}^1$ finite elements.
- 4 Compute now the solution using the following pairs of finite element spaces

$$\mathbb{P}^1/\mathbb{P}^0, \quad \mathbb{P}^1/\mathbb{P}^1, \quad \mathbb{P}_b^1/\mathbb{P}^1, \quad \mathbb{P}^2/\mathbb{P}^1.$$

Which of these pairs produce spurious pressure modes?

- 5 Use the results of the previous point to fill the following table: what can you observe?

N_{dofs}	$\ \mathbf{u} - \mathbf{u}_h\ _{L^2}$	$\ \nabla \mathbf{u} - \nabla \mathbf{u}_h\ _{L^2}$	$\ p - p_h\ _{L^2}$
$\mathbb{P}^1/\mathbb{P}^0$			
$\mathbb{P}^1/\mathbb{P}^1$			
$\mathbb{P}_b^1/\mathbb{P}^1$			
$\mathbb{P}^2/\mathbb{P}^1$			

Exercise 2

Let us consider the following Stokes problem over the domain $\Omega = (0, 2) \times (-1, 1)$:

$$\begin{cases} -\mu \Delta \mathbf{u} + \nabla p = \mathbf{0}, & \text{in } \Omega, \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, \\ \mathbf{u} = \mathbf{0}, & \text{on } \Gamma_{\text{wall}} = \{0 \leq x \leq 2, y \in \{-1, 1\}\}, \\ \mathbf{u} = (1 - y^2)\mathbf{i} + 0\mathbf{j}, & \text{on } \Gamma_{\text{in}} = \{x = 0, -1 \leq y \leq 1\}, \\ (\nabla \mathbf{u} - pI)\mathbf{n} = \mathbf{0}, & \text{on } \Gamma_{\text{out}} = \{x = 2, -1 \leq y \leq 1\}, \end{cases}$$

with $\mu = 10^{-2}$.

- 1 Write the weak formulation of the differential problem, strongly enforcing Dirichlet boundary conditions.
- 2 Introduce the finite element method for the problem at hand, using $\mathbb{P}^2$ elements for velocity and $\mathbb{P}^1$ elements for pressure, and derive its algebraic formulation.
- 3 Assemble the algebraic problem in Firedrake, using `LinearVariationalProblem`. Recall that the monolithic system has the following form:

$$\Sigma \mathbf{x} = \begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \mathbf{P} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{0} \end{bmatrix} = \mathbf{b}.$$

where A is the stiffness matrix, B and B^T correspond to the divergence and gradient operators, respectively, and $\mathbf{F}$ contains the boundary data.

- 4 Solve the algebraic problem assembled at the previous point using `LinearVariationalSolver`: use the direct solver `mumps` on the linear system, by setting the parameters `{'ksp_type': 'preonly', 'pc_type': 'lu', 'pc_factor_mat_solver_type': 'mumps'}`.
- 5 (additional) Repeat the exercise for different values of μ : what can you observe in the solution?